

The Icosian Closure Object as a Unification of Geometry and Arithmetic

(VFD programme — working draft)

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One sentence. A single geometric object — the icosian closure structure on the 600-cell — produces *both* its geometry (the E_8 root system, exact) *and* its arithmetic (the L -function $\zeta_K(s)\zeta_K(s-1)$, exact), and the *same* closure form runs through both; the Riemann witness is the one open frontier. **This is a unification of two domains in one object, both sides proved** — not an analogy, and not a proof of the Riemann Hypothesis (RH(L)) is the frontier, explicitly open; it is GRH for one cuspidal L , not classical RH).

Abstract

We argue that geometry and arithmetic meet, concretely and provably, inside one object. The icosian maximal order $\mathcal{I}$ on the 600-cell over $K = \mathbb{Q}(\sqrt{5})$ has two exact faces. *Geometrically*, its short vectors are the 240 roots of E_8 — once the naive embedding is completed to the golden trace form. *Arithmetically*, its theta series has L -function $\zeta_K(s)\zeta_K(s-1)$ exactly, with no local correction. The bridge between the faces is a single fact — positivity and self-adjointness are *form-relative* — and the same completed form that closes the geometry (E_8 even-integral) is the form in which the arithmetic positivity lives (the Weil form). On the arithmetic face this positivity is a self-adjoint operator A with $A \geq 0 \iff$ RH for the object's cuspidal L -function; that positivity is the open frontier. We make no claim about RH; the contribution is the *object* as a concrete meeting point of geometry and arithmetic, with the frontier named.

1 One object

The object is the *icosian closure structure* on the 600-cell: its 120 vertices are the unit icosians (the binary icosahedral group $2I$), generating the icosian ring $\mathcal{I}$, a maximal order in the definite quaternion algebra over $K = \mathbb{Q}(\sqrt{5})$ ramified at the two infinite places, of class number one, with underlying $\mathbb{Z}$ -lattice E_8 . It is rigid geometry — no parameters. The claim of this paper is that this *one* object carries two exact arithmetic/geometric structures, joined by one form (Fig. 1).

one object, two exact faces, one form -- and one open frontier ($A \geq 0 \Leftrightarrow \text{RH}(L)$)

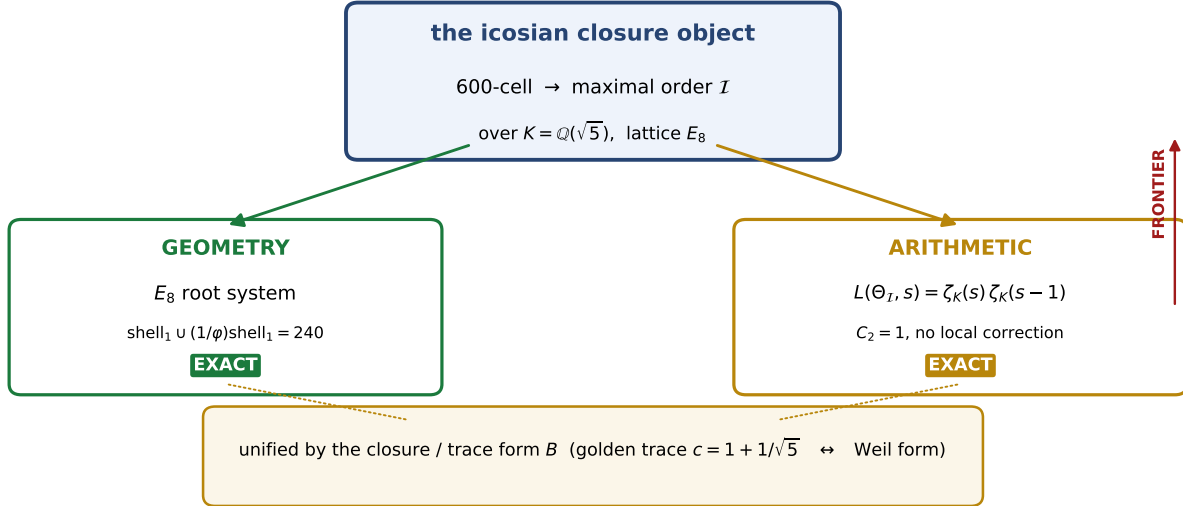


Figure 1: One object, two exact faces, one form. The icosian closure object yields the E_8 root system (geometry, exact) and the L -function $\zeta_K(s)\zeta_K(s-1)$ (arithmetic, exact); both are organised by the same closure/trace form B . The witness operator $A \geq 0 \Leftrightarrow \text{RH}(L)$ is the single open frontier.

2 The geometric face: E_8 , exactly

The naive embedding $x \mapsto (x, \sigma x)$ of the icosian lattice gives inner products $\pm \frac{1}{2}$ — not even-integral, so not a root lattice. Completing to the *golden trace form*

$$B(x, y) = \text{Tr}_{K/\mathbb{Q}}(c \langle x, y \rangle), \quad c = 1 + \frac{1}{\sqrt{5}} \quad (\text{Tr} c = 2),$$

with c forced (not tuned) by the norm-2 and integrality conditions, the two shells $\text{shell}_1 \cup (1/\varphi)\text{shell}_1$ become the 240 vectors of norm 2 with integer inner products, root graph $(56, 56, 126, 1, 1)$, rank 8, even integral: the canonical E_8 (Fig. 2). The residual closes $\pm \frac{1}{2} \rightarrow 0$. This is the known Wilson / Moody–Patera construction [1, 2], verified; we use it as the *geometric face* of the object.

geometry face (EXACT): 240 E_8 roots, Coxeter plane
 8 rings $\times$ 30, $\text{norm}^2 = 2$, even integral

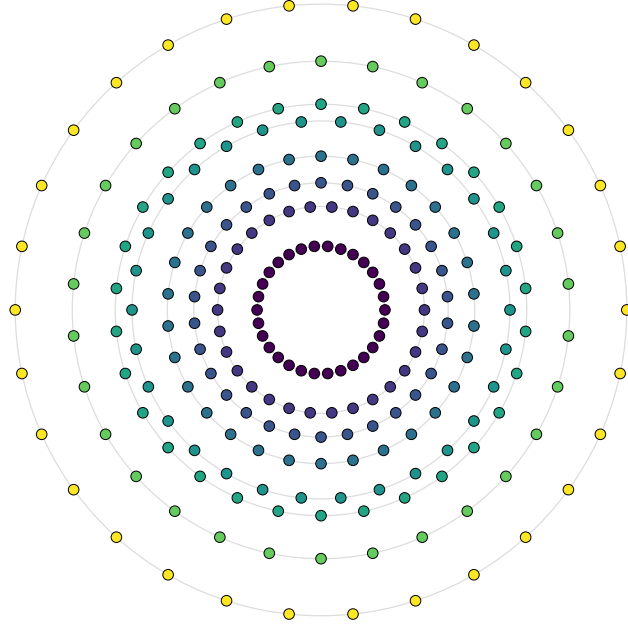


Figure 2: The geometric face (exact): the 240 E_8 roots in the Coxeter plane, eight rings of thirty (verified $240 = 8 \times 30$). The golden trace form makes the object's short vectors even-integral E_8 .

3 The arithmetic face: $\zeta_K(s)\zeta_K(s-1)$, exactly

The *same* order $\mathcal{I}$ has a theta series, and its L -function is the rank-two Dedekind product

$$L(\Theta_{\mathcal{I}}, s) = \zeta_K(s) \zeta_K(s-1) \quad \text{exactly,} \quad C_2 = 1,$$

with no local correction, by the Siegel–Weil/maximal-order theorem and the single ideal class [3]. Via $\zeta_K = \zeta \cdot L(s, \chi_5)$ the Riemann ζ appears as one of four factors (Fig. 3); the object *contains* ζ but does *not* single it out — a structurally certified negative [4]. The maximal-order representation numbers $r(\pi) = 120(1 + N(\pi))$ at every prime (inert 2: $r(2) = 600 = 120 \cdot 5$, no anomaly) give the standard Eisenstein local factor everywhere.

$$\begin{aligned}
L(\Theta_I, s) &= \zeta_K(s) \zeta_K(s-1) \text{quad}(\text{exact}, C_2 = 1) \\
&= \zeta(s) \cdot L(s, \chi_5) \cdot \zeta(s-1) \cdot L(s-1, \chi_5)
\end{aligned}$$

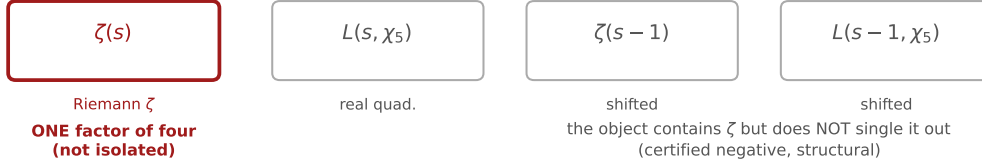


Figure 3: The arithmetic face (exact): the object's L -function factors as $\zeta_K(s)\zeta_K(s-1) = \zeta(s)L(s, \chi_5)\zeta(s-1)L(s-1, \chi_5)$. The Riemann ζ is one factor of four, not isolated.

4 One form joins the two faces

The two faces are not glued by hand; they are two readings of one form. The enabling fact:

Lemma 1 (form-relativity). *For an invertible symmetric form B and operator T , T is B -self-adjoint iff BT is symmetric (then $T^{\dagger_B} = B^{-1}T^{\top}B$).*

Multiplication by φ on K in the basis $\{1, \sqrt{5}\}$ has matrix $T = \begin{bmatrix} 1/2 & 5/2 \\ 1/2 & 1/2 \end{bmatrix}$ (not symmetric); under the trace form $B = \text{diag}(2, 10)$, $BT = \begin{bmatrix} 1 & 5 \\ 5 & 5 \end{bmatrix}$ is symmetric, so φ is B -self-adjoint (Fig. 4). The point: the *same* trace form B that makes the geometry close (E_8 even-integral, §2) is the form in which the arithmetic's self-adjointness and positivity are stated (§5). Geometry and arithmetic share not a metaphor but a form.

the shared form: multiplication by φ on $\mathbb{Q}(\sqrt{5})$

naive T $\begin{bmatrix} 1/2 & 5/2 \\ 1/2 & 1/2 \end{bmatrix}$	trace form $B = \text{diag}(2, 10)$ $B \ T = \begin{bmatrix} 1 & 5 \\ 5 & 5 \end{bmatrix}$
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$$T \neq T^{\top}$$

$$BT = (BT)^{\top} \Rightarrow B\text{-self-adjoint}$$

same operator: non-symmetric naively, self-adjoint in the trace form
self-adjointness / positivity is FORM-RELATIVE

Figure 4: The shared form (exact). Self-adjointness is form-relative: multiplication by φ is non-symmetric naively and self-adjoint in the trace form. The same completion closes the geometry and houses the arithmetic positivity.

5 The frontier: a witness operator, $A \geq 0 \iff \text{RH}$

On the arithmetic face the positivity becomes a single self-adjoint operator on the scale axis. With A_{∞} the archimedean boundary (the gamma factor, Tate's local theory) and A_P the prime residue

built from the object's geometric Brandt eigenvalues a_q ,

$$A = A_\infty - A_P, \quad \langle h, Ah \rangle = \sum_{\rho} |\hat{h}(\gamma_\rho)|^2, \quad \boxed{A \geq 0 \iff \text{RH}(L)}$$

(construction, identification, and numerical corroboration in [5]; Fig. 5). The operator is self-adjoint, and its form reproduces the known Riemann zeros to $\sim 10^{-7}$; *that* $A \geq 0$ *holds* is exactly RH for the object's cuspidal L -function (GRH for one L , not classical RH), and it is open. This is the single frontier of the picture: everything else — the geometry, the arithmetic identity, the shared form — is exact and behind us; the positivity is the wall ahead.

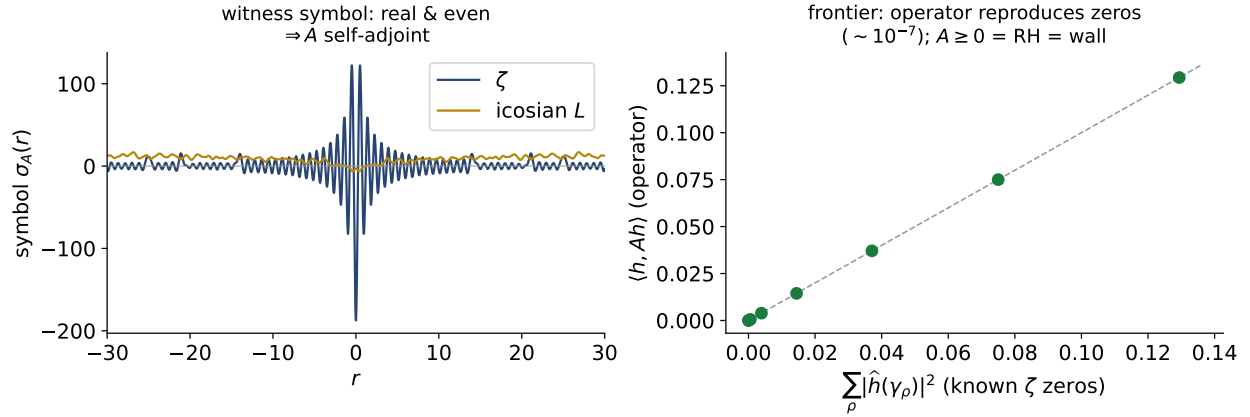


Figure 5: The frontier. Left: the witness symbol $\sigma_A(r)$ for ζ and the object's L is real and even, so A is self-adjoint. Right: the operator form reproduces the sum over the known Riemann zeros ($\sim 10^{-7}$). Positivity of A for all h is $\text{RH}(L)$ — the open frontier, not shown.

6 What this unifies, and a noted echo

The unification. Geometry and arithmetic are not analogous here; they are two exact faces of one concrete object, joined by one form. The object's short vectors *are* E_8 ; its theta series *is* $\zeta_K(s)\zeta_K(s-1)$; the trace form that closes the first is the form that houses the positivity of the second. Both faces are proved; only the positivity frontier is open.

A noted echo (not a rung). The same golden/ σ -paired form reappears in a separate cosmological *model* [6]. We record this only as a recurrence of the same form-language in another domain; we make *no* claim that the physics implies or is implied by the arithmetic, and cosmology and RH never touch as a claim.

What is not claimed. No proof of RH or GRH; the positivity frontier is open. The arithmetic face is the *cuspidal* L for the witness (GRH for one L); classical ζ enters only as one factor of the Eisenstein face and is not isolated. The contribution is the object as a concrete meeting point of geometry and arithmetic, with the one open step named precisely.

References

- [1] R. A. Wilson, icosian / E_8 constructions (quaternionic reflection groups).
- [2] R. V. Moody, J. Patera, *Quasicrystals and icosians*, J. Phys. A **26** (1993).

- [3] VFD programme, *The Icosian Triad on V_{600}* (icosian-triad-v600 v1.1): $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\zeta_K(s-1)$, $C_2 = 1$, exact.
- [4] VFD programme, *A Closure Geometry on the 600-Cell and Its Arithmetic Shadow* (icosian-closure-object): the certified ζ -boundary.
- [5] VFD programme, *A Witness Operator on the Scale Axis* (companion): $A = A_\infty - A_P$, $A \geq 0 \iff \text{RH}(L)$; geometric coefficients; numerical corroboration.
- [6] VFD programme, *Hypersphere Cosmology* (v1.1): the σ -paired form in a cosmological model (separate domain; firewalled from RH).
- [7] VFD programme, *Invariant Trace-Form Closure* (internal report): the form-relativity lens and its grading.